Class18

Matthew Huang (PID A17978767)

Background

Pertusis (a.k.a. whooping cough) is a common lung infection caused by bacteria B. Pertusis.

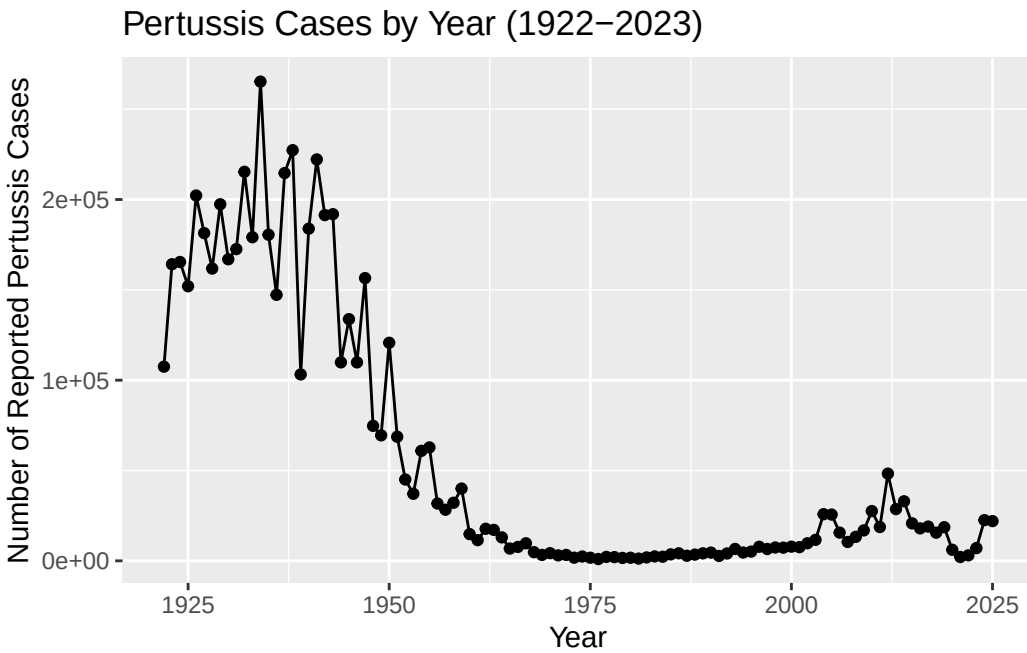
The CDC tracks the number of reported data in the US: We can "scrape" this data with the **datapasta** package.

```
cdc <- data.frame(data.table::data.table(
                                         year = c(1922L,1923L,
                                                  1924L,1925L,1926L,1927L,1928L,
                                                  1929L,1930L,1931L,1932L,1933L,
                                                  1934L,1935L,1936L,1937L,
                                                  1938L,1939L,1940L,1941L,1942L,
                                                  1943L,1944L,1945L,1946L,1947L,
                                                  1948L,1949L,1950L,1951L,
                                                  1952L,1953L,1954L,1955L,1956L,
                                                  1957L,1958L,1959L,1960L,1961L,
                                                  1962L,1963L,1964L,1965L,1966L,
                                                  1967L,1968L,1969L,1970L,
                                                  1971L,1972L,1973L,1974L,1975L,
                                                  1976L,1977L,1978L,1979L,1980L,
                                                  1981L,1982L,1983L,1984L,
                                                  1985L,1986L,1987L,1988L,1989L,
                                                  1990L,1991L,1992L,1993L,1994L,
                                                  1995L,1996L,1997L,1998L,1999L,
                                                  2000L,2001L,2002L,2003L,
                                                  2004L,2005L,2006L,2007L,2008L,
                                                  2009L,2010L,2011L,2012L,2013L,
                                                  2014L,2015L,2016L,2017L,
                                                  2018L,2019L,2020L,2021L,2022L,
                                                  2023L,2024L,2025L),
                cases = c(107473,164191,
```

```
                                                    165418,152003,202210,181411,
                                                    161799,197371,166914,172559,
                                                    215343,179135,265269,180518,
                                                    147237,214652,227319,103188,
                                                    183866,222202,191383,191890,
                                                    109873,133792,109860,156517,74715,
                                                    69479,120718,68687,45030,
                                                    37129,60886,62786,31732,28295,
                                                    32148,40005,14809,11468,17749,
                                                    17135,13005,6799,7717,9718,
                                                    4810,3285,4249,3036,3287,
                                                    1759,2402,1738,1010,2177,2063,
                                                    1623,1730,1248,1895,2463,
                                                    2276,3589,4195,2823,3450,4157,
                                                    4570,2719,4083,6586,4617,
                                                    5137,7796,6564,7405,7298,7867,
                                                    7580,9771,11647,25827,25616,
                                                    15632,10454,13278,16858,27550,
                                                    18719,48277,28639,32971,
                                                    20762,17972,18975,15609,18617,
                                                    6124,2116,3044,7063,22538,21996)
            ))
```

Q. Make a plot of `year` vs `cases`

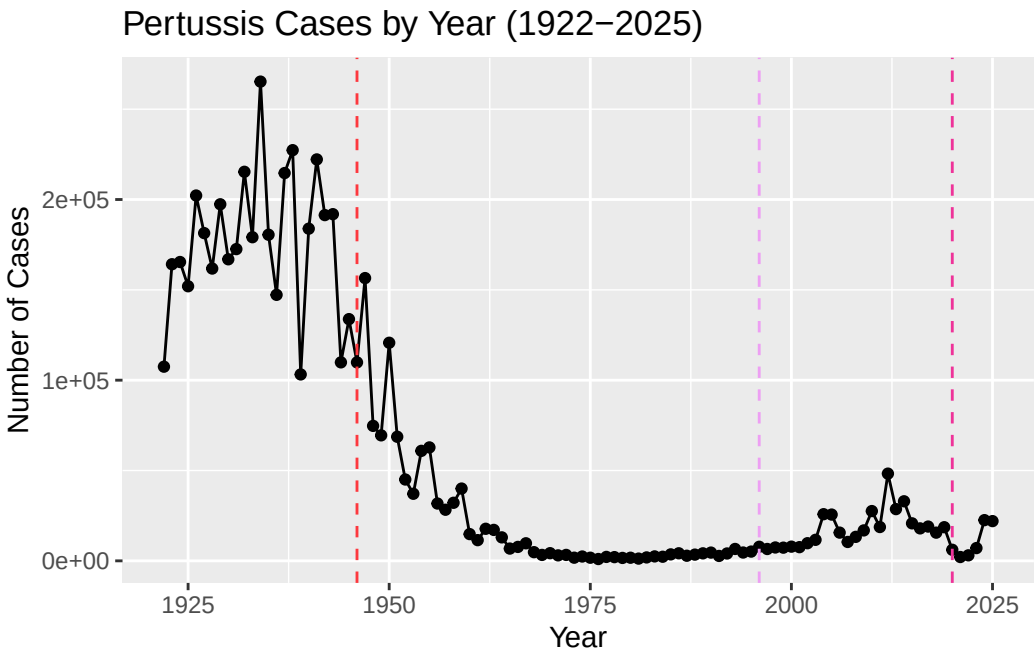
```
library(ggplot2)
ggplot(cdc) +
  aes(year, cases) +
  geom_line() +
  geom_point() +
  labs(x = "Year", y = "Number of Reported Pertussis Cases", title = "Pertussis Cases by Yea
```



Q2. Using the ggplot `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

```
ggplot(cdc) +
  aes(year, cases) +
  geom_point() +
  geom_line() +

  geom_vline(xintercept = 1946, col = "#FD383e", lty=2) +
  geom_vline(xintercept = 1996, col = "#ed9ff3", lty=2) +
  geom_vline(xintercept = 2020, col = "#e39", lty=2) +
  labs(x = "Year", y = "Number of Cases", title = "Pertussis Cases by Year (1922-2025)")
```



There was a high number of cases before the wP vaccine and then a decline and stabilization, followed by an increase in 1996 after the introduction of the aP vaccine. Relatively stable from 1975 to 1996 and then unstable again.

> Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

In the 1940s, pertussis cases were very high, but the introduction of the whole-cell pertussis vaccine wP in 1946 led to a major decline in disease incidence. However, after the acellular pertussis vaccine aP replaced wP in 1996, cases began to increase again. One likely explanation is that protection from the aP vaccine wanes faster over time than protection from wP. In addition, the rise in reported cases during the 2000s may also reflect more sensitive PCR-based testing, increased vaccine hesitancy, and genetic changes in Bordetella pertussis that may help the bacterium evade vaccine-induced immunity.

CMI-PB project

Computational Models of Immunity - Pertussis Boost project aims to provide the scienntific community with this very information.

They make there data available via JSON format returnig API. We ca read this in R with the `read_json()` function from the **jsonlite** package:

```r
library(jsonlite)

subject <- read_json("http://cmi-pb.org/api/v5_1/subject", simplifyVector = TRUE)
head(subject)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          2          wP         Female Not Hispanic or Latino White
3          3          wP         Female                Unknown White
4          4          wP           Male Not Hispanic or Latino Asian
5          5          wP           Male Not Hispanic or Latino Asian
6          6          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset
1    1986-01-01    2016-09-12 2020_dataset
2    1968-01-01    2019-01-28 2020_dataset
3    1983-01-01    2016-10-10 2020_dataset
4    1988-01-01    2016-08-29 2020_dataset
5    1991-01-01    2016-08-29 2020_dataset
6    1988-01-01    2016-10-10 2020_dataset
```

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

```r
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q5. How many Male and Female subjects/patients are in the dataset?

```r
table(subject$biological_sex)
```

```
Female   Male
   112     60
```

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc…)?

```r
table(subject$race, subject$biological_sex)
```

```
                                          Female Male
  American Indian/Alaska Native                0    1
  Asian                                       32   12
  Black or African American                    2    3
  More Than One Race                          15    4
  Native Hawaiian or Other Pacific Islander    1    1
  Unknown or Not Reported                     14    7
  White                                       48   32
```

Let's read some more database tables:

```r
specimen <- read_json("http://cmi-pb.org/api/v5_1/specimen", simplifyVector = TRUE)
ab_titer <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector = TRUE)
```

```r
head(specimen)
```

```
  specimen_id subject_id actual_day_relative_to_boost
1           1          1                           -3
2           2          1                            1
3           3          1                            3
4           4          1                            7
5           5          1                           11
6           6          1                           32
  planned_day_relative_to_boost specimen_type visit
1                             0         Blood     1
2                             1         Blood     2
3                             3         Blood     3
4                             7         Blood     4
5                            14         Blood     5
6                            30         Blood     6
```

```r
head(ab_titer)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
```

```
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection
1 UG/ML                 2.096133
2 IU/ML                29.170000
3 IU/ML                 0.530000
4 IU/ML                 6.205949
5 IU/ML                 4.679535
6 IU/ML                 2.816431
```

To analyze this data we need to first "join" (merge/link) the different tables so we have all the data inn one place not spread across different tables.

We can use the `*_join()` family of functions from **dplyr** to do this

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
meta<- inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```r
head(meta)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
```

```
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           2
3    1986-01-01    2016-09-12 2020_dataset           3
4    1986-01-01    2016-09-12 2020_dataset           4
5    1986-01-01    2016-09-12 2020_dataset           5
6    1986-01-01    2016-09-12 2020_dataset           6
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                            1                             1         Blood
3                            3                             3         Blood
4                            7                             7         Blood
5                           11                            14         Blood
6                           32                            30         Blood
  visit
1     1
2     2
3     3
4     4
5     5
6     6
```

```r
abdata <- inner_join(ab_titer, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```r
head(abdata)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 UG/ML                 2.096133          1          wP         Female
2 IU/ML                29.170000          1          wP         Female
```

```
3 IU/ML                 0.530000          1          wP         Female
4 IU/ML                 6.205949          1          wP         Female
5 IU/ML                 4.679535          1          wP         Female
6 IU/ML                 2.816431          1          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

```r
library(dplyr)
meta <- inner_join(subject, specimen)
```

```
Joining with `by = join_by(subject_id)`
```

```r
head(meta)
```

```
  subject_id infancy_vac biological_sex              ethnicity  race
1          1          wP         Female Not Hispanic or Latino White
2          1          wP         Female Not Hispanic or Latino White
3          1          wP         Female Not Hispanic or Latino White
```

```
4          1          wP         Female Not Hispanic or Latino White
5          1          wP         Female Not Hispanic or Latino White
6          1          wP         Female Not Hispanic or Latino White
  year_of_birth date_of_boost      dataset specimen_id
1    1986-01-01    2016-09-12 2020_dataset           1
2    1986-01-01    2016-09-12 2020_dataset           2
3    1986-01-01    2016-09-12 2020_dataset           3
4    1986-01-01    2016-09-12 2020_dataset           4
5    1986-01-01    2016-09-12 2020_dataset           5
6    1986-01-01    2016-09-12 2020_dataset           6
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                            1                             1         Blood
3                            3                             3         Blood
4                            7                             7         Blood
5                           11                            14         Blood
6                           32                            30         Blood
  visit
1     1
2     2
3     3
4     4
5     5
6     6
```

Q10. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```r
abdata <- inner_join(ab_titer, meta)
```

```
Joining with `by = join_by(specimen_id)`
```

```r
head(abdata)
```

```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgE               FALSE   Total 1110.21154       2.493425
2           1     IgE               FALSE   Total 2708.91616       2.493425
3           1     IgG                TRUE      PT   68.56614       3.736992
4           1     IgG                TRUE     PRN  332.12718       2.602350
5           1     IgG                TRUE     FHA 1887.12263      34.050956
6           1     IgE                TRUE     ACT    0.10000       1.000000
```

```
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 UG/ML                 2.096133          1          wP         Female
2 IU/ML                29.170000          1          wP         Female
3 IU/ML                 0.530000          1          wP         Female
4 IU/ML                 6.205949          1          wP         Female
5 IU/ML                 4.679535          1          wP         Female
6 IU/ML                 2.816431          1          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
5 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```r
table(abdata$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4 
 6698  7265 11993 12000 12000 12000 
```

What antigens are reported?

```r
table(abdata$antigen)
```

```
    ACT   BETV1      DT   FELD1     FHA  FIM2/3   LOLP1     LOS Measles     OVA
   1970    1970    6318    1970    6712    6318    1970    1970    1970    6318
    PD1     PRN      PT     PTM   Total      TT
   1970    6712    6712    1970     788    6318
```

Q12. What are the different $dataset values in abdata and what do you notice about the number of rows for the most "recent" dataset?

```r
table(abdata$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
       31520         8085         7301        15050
```

The number of rows in the most recent dataset is much higher than in either of the previous two years individually. It is slightly lower than the combined total of the 2021 and 2022 datasets, but about half the size of the 2020 dataset.

Examine IgB Ab titer levels

```r
igg <- abdata |>
  filter(isotype == "IgG")
head(igg)
```

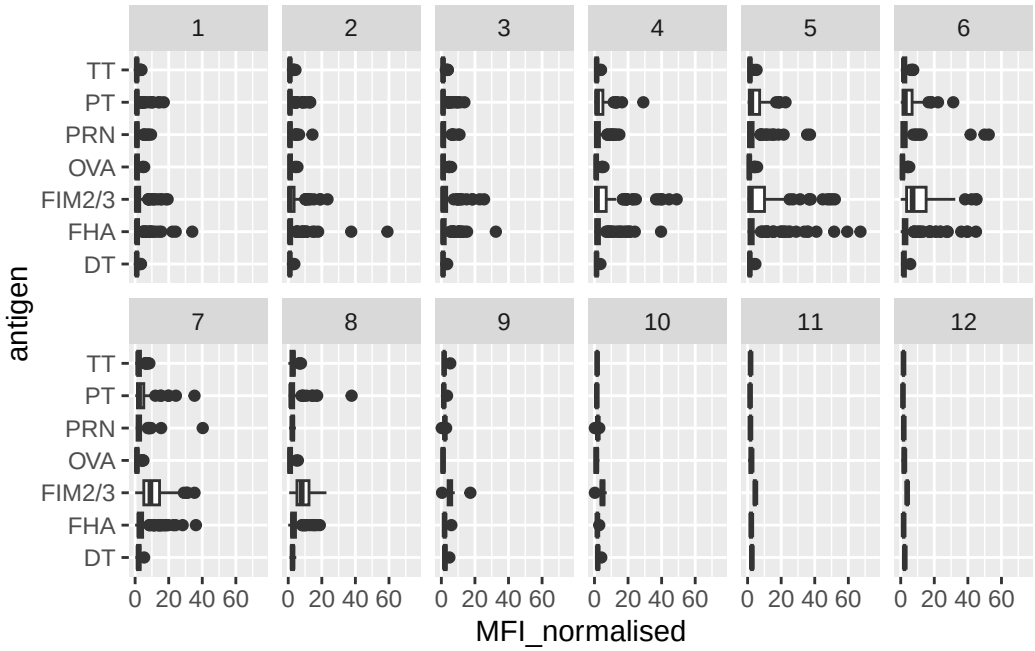
```
  specimen_id isotype is_antigen_specific antigen        MFI MFI_normalised
1           1     IgG                TRUE      PT   68.56614       3.736992
2           1     IgG                TRUE     PRN  332.12718       2.602350
3           1     IgG                TRUE     FHA 1887.12263      34.050956
4          19     IgG                TRUE      PT   20.11607       1.096366
5          19     IgG                TRUE     PRN  976.67419       7.652635
6          19     IgG                TRUE     FHA   60.76626       1.096457
   unit lower_limit_of_detection subject_id infancy_vac biological_sex
1 IU/ML                 0.530000          1          wP         Female
2 IU/ML                 6.205949          1          wP         Female
3 IU/ML                 4.679535          1          wP         Female
4 IU/ML                 0.530000          3          wP         Female
5 IU/ML                 6.205949          3          wP         Female
6 IU/ML                 4.679535          3          wP         Female
               ethnicity  race year_of_birth date_of_boost      dataset
```

```
1 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
2 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
3 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
4                Unknown White    1983-01-01    2016-10-10 2020_dataset
5                Unknown White    1983-01-01    2016-10-10 2020_dataset
6                Unknown White    1983-01-01    2016-10-10 2020_dataset
  actual_day_relative_to_boost planned_day_relative_to_boost specimen_type
1                           -3                             0         Blood
2                           -3                             0         Blood
3                           -3                             0         Blood
4                           -3                             0         Blood
5                           -3                             0         Blood
6                           -3                             0         Blood
  visit
1     1
2     1
3     1
4     1
5     1
6     1
```

Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

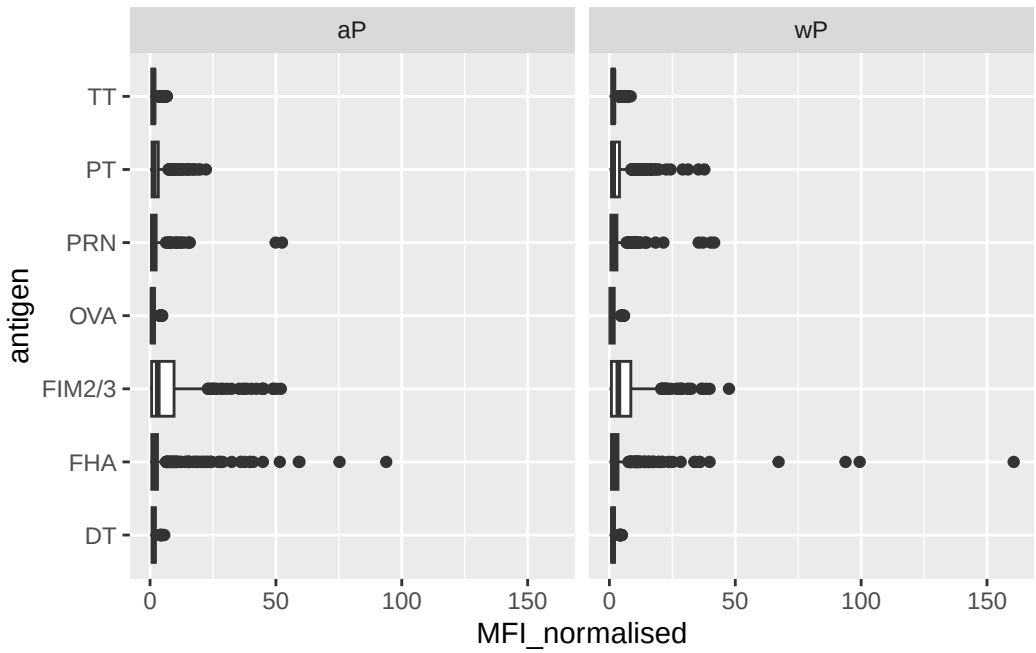
```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot()  +
   xlim(0,75) +
  facet_wrap(vars(visit), nrow=2)
```

```
Warning: Removed 5 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```

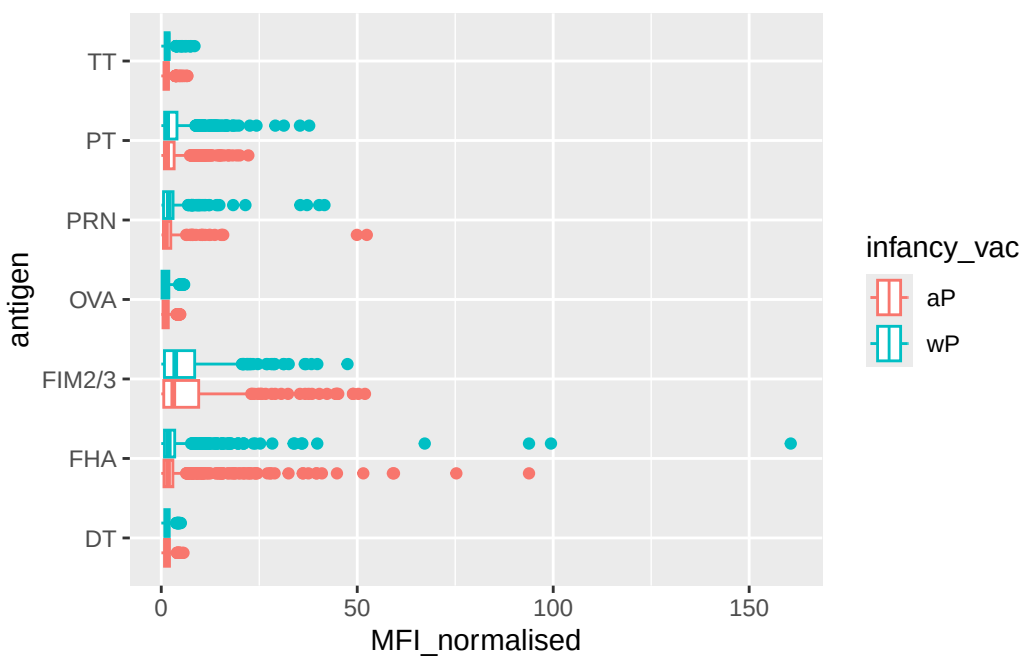


Q. Is there a difference between aP vs wP vaccinated individuals with these values?

```
ggplot(igg) +
  aes(MFI_normalised, antigen) +
  geom_boxplot() +
  facet_wrap(~infancy_vac)
```



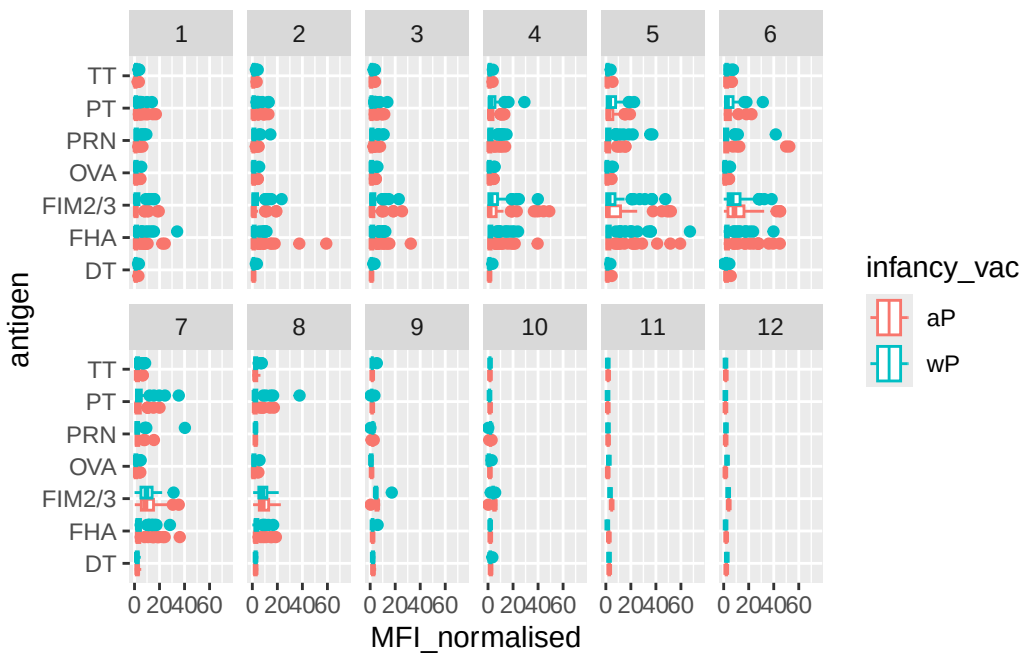

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot()
```



Q. Is there a temprol response - i.e do values increase or decrease over time?

```
ggplot(igg) +
  aes(MFI_normalised, antigen, col=infancy_vac) +
  geom_boxplot() +
  xlim(0,75) +
  facet_wrap(vars(visit), nrow=2)
```

```
Warning: Removed 5 rows containing non-finite outside the scale range
(`stat_boxplot()`).
```



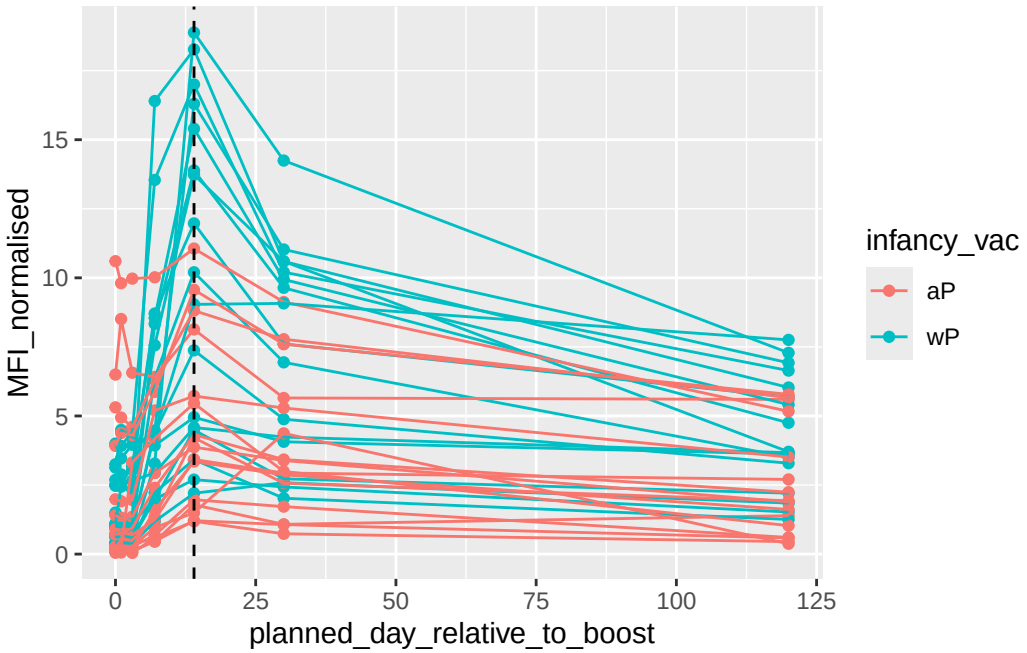
The values tend to initially increase and then decrease as more time passes.

Focus on "PT" Pertussis Toxin antigen

```
pt.igg.21 <- igg |> filter(antigen == "PT",
               dataset == "2021_dataset")

ggplot(pt.igg.21) +
  aes(planned_day_relative_to_boost,
```

```
    MFI_normalised,
    col=infancy_vac,
    group = subject_id)+
  geom_point() +
  geom_line() +
  geom_vline(xintercept = 14, lty=2)
```



Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

The antigens that show the clearest changes in IgG antibody titers over time are PT and FIM 2/3, since their boxplots become more prominent across visits and show higher median MFI values at the middle time points. PRN and FHA also show some increase in the middle visits, but the pattern is less pronounced. In contrast, TT, OVA, and DT show the weakest changes over time. This suggests that PT and FIM 2/3 may trigger stronger or more variable antibody responses than the other antigens.